

Method Overloading, Constructor Overloading and Object as Parameter

A. Method (Function) Overloading

1. Write a Java program to create a Calculator class with overloaded add() methods to add two integers, three integers, and two double values.
2. Write a Java program to create an Area class with overloaded calculateArea() methods to calculate the area of a square, rectangle, and circle.
3. Write a Java program to create a Maximum class with overloaded max() methods to find the maximum of two integers, three integers, and two double values.
4. Write a Java program to create a Student class with overloaded display() methods to display a student's name, name with marks, and name with marks and grade.
5. Write a Java program to create a Printer class with overloaded print() methods to print an integer, a double value, a character, and a String.
6. Write a Java program to create a Number class with overloaded multiply() methods to multiply two integers, three integers, and two double values.

B. Constructor Overloading

7. Write a Java program to create a Student class with overloaded constructors to initialize only the name, name and roll number, and name, roll number, and marks.
8. Write a Java program to create an Employee class with overloaded constructors to initialize employee ID, employee ID and name, and employee ID, name, and salary.
9. Write a Java program to create a Rectangle class with overloaded constructors: a default constructor, a constructor accepting one value for a square, and a constructor accepting length and breadth for a rectangle. Display the area.
10. Write a Java program to create a Box class with overloaded constructors to initialize no dimensions, one dimension, and length, width, and height. Calculate and display the volume.
11. Write a Java program to demonstrate constructor overloading in a Book class using constructors with no arguments, title only, and title with author and price.
12. Write a Java program to demonstrate constructor chaining using this() with three overloaded constructors in a Student class.

C. Object as a Parameter

13. Write a Java program to create a Student class containing roll number and marks. Create a compare(Student s) method to compare the marks of two Student objects.
14. Write a Java program to create an Employee class containing ID, name, and salary. Create a compareSalary(Employee e) method to compare the salaries of two Employee objects.
15. Write a Java program to create a Distance class containing feet and inches. Create an add(Distance d) method that accepts another Distance object and returns a new Distance object containing their sum.

16. Write a Java program to create a Number class containing an integer value. Create an add(Number n) method that adds the value of another Number object to the current object.
17. Write a Java program to create a Time class containing hours and minutes. Create an add(Time t) method to add two Time objects and display the result in hours and minutes.
18. Write a Java program to create a Complex class containing real and imaginary parts. Create an add(Complex c) method to add two complex number objects.

D. Combined Programming Questions

19. Write a Java program to create a Box class using constructor overloading. Also create a compare(Box b) method that compares the volume of the current Box object with another Box object.
20. Write a Java program to create a Calculator class with overloaded add() methods for integers, double values, and two Number objects.
21. Write a Java program to create a BankAccount class using constructor overloading and a transfer(BankAccount b, double amount) method to transfer money from one account object to another.
22. Write a Java program to create a Product class with overloaded constructors for different initialization options. Create a comparePrice(Product p) method to compare the prices of two products.
23. Write a Java program to create a Rectangle class with overloaded constructors and a compareArea(Rectangle r) method that compares the area of two Rectangle objects.
24. Write a Java program demonstrating method overloading, constructor overloading, and passing an object as a parameter using a single real-life example of your choice.